

1. (15%) 解聯立方程組(以列運算化簡增廣矩陣)

$$x_1 + 2x_2 - x_3 + 3x_4 = 3$$

$$x_1 + x_2 + 3x_3 + 2x_4 = -2$$

$$2x_1 - x_2 + 4x_3 + 3x_4 = 1$$

$$2x_1 - 2x_2 + 8x_3 + 6x_4 = -4$$

2. (14%) 矩陣 A 如右, 先以列運算求 A^{-1} , 並將 A 表示成一些基本矩陣的乘積:

$$\begin{bmatrix} 2 & 1 & 4 \\ 3 & 2 & 5 \\ 0 & -1 & 1 \end{bmatrix}$$

3. (10%) 向量 $[1, 2, 3, 7]$ 是否在 $\text{sp}([1, 0, 1, -3], [0, -1, 0, 0], [1, -3, -1, 0], [-1, 4, 2, -1])$ 中?

4. (10%) 令矩陣 A 如題 2, 求一矩陣 B 滿足 $AB = -A^2 + 3A$.

5. (15%) 矩陣 A 如右: 求 (1) $\text{rank}(A)$. (2) A 之 column space 的基底. (3) A 之 nullspace 的基底

$$\begin{bmatrix} 1 & 2 & -5 & 0 & -1 \\ 2 & 5 & -8 & 4 & 3 \\ -3 & -9 & 9 & -7 & -2 \end{bmatrix}$$

6. (14%) 是非題 (認為非者需說明)

- () a. If A is invertible, then $A + A^T$ is invertible.
- () b. Every matrix is row equivalent to a unique matrix in reduced row-echelon form.
- () c. A linear system with fewer equations than unknowns may have no solutions.
- () d. If $AB = C$ and two of the matrices are singular, then so is the third.
- () e. There are exactly two unit vectors perpendicular to any given nonzero vector in $\mathbb{R}^n$.
- () f. If A and B are square matrices, then $(A + B)^2 = A^2 + 2AB + B^2$.
- () g. The number of independent row vectors in a matrix is the same as the independent columns vectors.

7. (12%) 若 A, C 為同大小的方陣,

(a) 說明: $\text{rank}(AC) \leq \text{rank}(A)$

(b) 舉一例說明: $\text{rank}(AC) < \text{rank}(C)$

8. (10%) Let $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3$ be any vectors in $\mathbb{R}^n$. Explain that

$\mathbf{w}_1 = \mathbf{v}_1 - 4\mathbf{v}_2 + 3\mathbf{v}_3, \mathbf{w}_2 = 3\mathbf{v}_1 - 11\mathbf{v}_2 + 2\mathbf{v}_3, \mathbf{w}_3 = \mathbf{v}_1 - 3\mathbf{v}_2 - 4\mathbf{v}_3$ are dependent.